

The LoRA certificate

1. What we want

Someone fine-tunes a LoRA adapter on a few private images and publishes the two factors A_T and B_T . We show that these matrices give linear equations that every private feature must satisfy, and that the equations can be written down from the published factors alone, without the seed or training schedule. Call a matrix giving such equations a certificate. The equations act on features, so to search over images we first map each candidate to its features.

Consider one adapted linear layer $W_0 + B_tA_t$, with frozen $W_0 \in \mathbb{R}^{m \times n}$, $A_t \in \mathbb{R}^{r \times n}$, and $B_t \in \mathbb{R}^{m \times r}$. Here r is the adapter rank, and n, m are the input and output dimensions. We initialize at step 0 and release the factors at step T .

Notation. For N private images x_i , let $h_i = \phi(x_i) \in \mathbb{R}^n$ be their inputs to this layer and set $H = [h_1, \dots, h_N] \in \mathbb{R}^{n \times N}$, $q = \text{rank } H$. Here q counts independent feature directions, so $q \leq N$. Define

$$\mathcal{S} = \text{col}(A_0H), \quad P = \text{proj}_{\mathcal{S}^\perp}, \quad P_T = \text{proj}_{\text{row}(B_T)^\perp}.$$

Here proj denotes orthogonal projection in $\mathbb{R}^r$. $\mathcal{S}$ is the image of the private features under the seed A_0 . Call it the *excited subspace*, the only part of $\mathbb{R}^r$ touched by the SGD updates. We can compute P_T from the released B_T , while P appears only in the proof. Since the columns of A_0H lie in $\mathcal{S}$, we have $PA_0H = 0$.

The adapter starts as a random A_0 and a zero B_0 . Training can only move A in directions the private features excite, so its component outside those directions stays exactly as it was. Under excitation, the rows of B_T span the excited subspace and tell us which part was touched. Projecting A_T away from that subspace leaves a piece of the seed that vanishes on every private feature. That is the certificate. We prove that $P_T = P$, so $C = P_TA_T$ equals PA_0 , vanishes on H , and has rank $r - q$.

2. How it is used

We search a public image family, or *chart*, $x = G(z)$, where $z \in \mathbb{R}^k$ contains the k free coordinates of one candidate image (for example, its retained PCA coefficients). Write $h(z) = \phi(G(z))$.

Doubling h doubles Ch , so the size of Ch alone does not give a relative measure of fit. We normalize by the size of the transformed feature:

$$F(z) = \frac{Ch(z)}{\|A_T h(z)\|_2}, \quad \text{score}(z) = \|F(z)\|_2, \quad A_T h(z) \neq 0.$$

The score is the fraction of $A_T h$ that sticks out of $\text{row}(B_T)$. That fraction is a sine, lies in $[0, 1]$, and is unchanged by rescaling h .

Under our assumptions and in exact arithmetic, a positive score excludes a candidate. A zero score says its features satisfy the equations, which is compatibility, not membership. When $A_T h(z) = 0$, both numerator and denominator vanish, and we skip the candidate.

3. What we assume, and why

A1. Initialization. We set $B_0 = 0$ and draw the entries of A_0 independently from distributions with densities, independently of H .

Why: This is the usual LoRA initialization, and $B_0A_0 = 0$ starts training at the pretrained model. For $q \leq r \leq n$, it gives $\text{rank } A_0 = r$ and $\text{rank}(A_0H) = q$ almost surely. Each rank failure forces a determinant to vanish. That determinant is a polynomial in the entries of A_0 that is not identically zero, because the dimensions allow a matrix of the required rank. Its zero set has measure zero. The strict inequality $q < r$ is what leaves a nonzero complement for the certificate.

A2. Vanilla SGD. We update both factors using minibatches and scalar learning rates η_t , with a differentiable loss of the layer outputs. The proofs use no momentum, weight decay or other parameter regularization.

Why: The proof needs each update to be a scalar multiple of the gradient, so that gradients confined to a subspace produce updates confined to it. Minibatches and a varying learning rate do not change this.

Extensions. Momentum with zero initial buffers combines gradients within the same subspaces. Scalar weight decay rescales the preserved seed component, which is fine as long as that scale is nonzero at release. Adam and AdamW are not covered because their per-entry scaling does not preserve these subspaces in general.

A3. Fixed inputs. The known map ϕ gives the same feature h_i at every step, so H stays fixed.

Why: Fixed inputs let us follow each image’s contribution throughout training. This is a head adapter, or the first adapted layer behind a frozen deterministic backbone with fixed preprocessing.

A4. Few-shot dimensions. We assume $q < r \leq n$ and $q \leq m$. For a softmax logit layer with a loss depending only on probabilities, we require $q \leq m - 1$.

Why: We need a nonzero complement for the certificate. We have in mind a handful of images, each represented by one pooled feature vector from a frozen backbone, so $q \leq N$ and choosing $r > N$ leaves that complement. For concreteness, $q = 8$, $r = 16$ and $m = n = 512$ leave eight equations on a 512-dimensional feature. The chart turns these into equations on k image coordinates.

The bound $q \leq m$ allows q independent output-gradient directions to exist. A5 explains the bound $q \leq m - 1$ at a softmax layer.

A5. Excitation at release. We require $\text{rank } B_T = q$: the rows of B_T must span $\mathcal{S}$.

Why: Excitation makes the full excited subspace visible in B_T . The update to B uses $D_t(A_t H)^\top$, with D_t defined in Claim 1, so independent output gradients can fill the space in a single full-batch step. Concretely, suppose the private features are independent, so $q = N$, and the first step uses all images. If $D_0 \in \mathbb{R}^{m \times N}$ has independent columns and $\eta_0 \neq 0$, then

$$B_1 = -\eta_0 D_0 (A_0 H)^\top, \quad \text{rank } B_1 = N = q.$$

This update has rank q , since $(A_0 H)^\top$ has full row rank and D_0 has full column rank. The condition has to hold at release, because later steps can cancel what earlier ones wrote.

A4’s width bounds describe the room these gradients have. At a softmax logit layer with a loss depending only on probabilities, shifting all logits equally changes nothing, so every output-error column is orthogonal to the all-ones vector $\mathbf{1}$. SGD passes this to the columns of B_T , giving $\text{rank } B_T \leq m - 1$. At a hidden layer, the downstream network determines which output-gradient directions are available.

Without excitation the invariant still gives $\text{row}(B_T) \subseteq \mathcal{S}$, hence $s := \text{rank } B_T \leq q$. The release always gives a lower bound on the private feature rank. Under excitation the bound is tight, so q can be read off B_T . In practice a gap above the numerical noise level separates its active singular directions from noise. The condition matters because $s < q$ enlarges the projector, and the extra directions can reject true features.

4. Three claims

Throughout, we work on the almost-sure event of A1, where $\text{rank } A_0 = r$ and $\text{rank}(A_0 H) = q$.

Claim 1. The gradients. Write $Z_t = (W_0 + B_t A_t)H$, let L_t be the minibatch loss, and set $D_t = \partial L_t / \partial Z_t$, with zero columns for examples outside the batch. Then

$$\boxed{\nabla_B L_t = D_t (A_t H)^\top, \quad \nabla_A L_t = B_t^\top D_t H^\top.}$$

Proof. Chain rule on $Z = W_0 H + B(AH)$, entry by entry, with a, b, c, i the output, adapter, input and example indices. Since $Z_{ai} = (W_0 H)_{ai} + \sum_{b,c} B_{ab} A_{bc} H_{ci}$,

$$\begin{aligned} \frac{\partial L}{\partial B_{ab}} &= \sum_i \frac{\partial L}{\partial Z_{ai}} \frac{\partial Z_{ai}}{\partial B_{ab}} = \sum_i D_{ai} (AH)_{bi}, \\ \frac{\partial L}{\partial A_{bc}} &= \sum_{a,i} \frac{\partial L}{\partial Z_{ai}} \frac{\partial Z_{ai}}{\partial A_{bc}} = \sum_{a,i} D_{ai} B_{ab} H_{ci}. \end{aligned}$$

These two products are the only way the private features enter the updates. The invariant below comes from reading them.

Claim 2. The invariant. At every step,

$$\boxed{B_t P = 0, \quad P A_t = P A_0.}$$

Proof. At initialization, $B_0 P = 0$ because $B_0 = 0$, and $P A_0 = P A_0$. Suppose both identities hold at step t . Then $P A_t H = P A_0 H = 0$, and Claim 1 gives

$$\begin{aligned} (\nabla_B L_t) P &= D_t (P A_t H)^\top = 0, \\ P \nabla_A L_t &= (B_t P)^\top D_t H^\top = 0. \end{aligned}$$

The vanilla SGD updates therefore give

$$\begin{aligned} B_{t+1} P &= (B_t - \eta_t \nabla_B L_t) P = 0, \\ P A_{t+1} &= P (A_t - \eta_t \nabla_A L_t) = P A_t = P A_0. \end{aligned}$$

In words, the rows of B stay in the excited subspace, and A keeps what it started with outside that space. The two gradient formulas preserve both properties at every step.

Claim 3. The observable certificate. Under A1–A5, $P_T = P$ and

$$\boxed{C = P_T A_T = P A_0, \quad C H = 0, \quad \text{rank } C = r - q.}$$

Proof. A1 and A4 give $\dim \mathcal{S} = q$. Claim 2 gives $\text{row}(B_T) \subseteq \mathcal{S}$, and A5 makes the dimensions equal. Thus $P_T = P$ and $C = P_T A_T = P A_0$, so $C H = P A_0 H = 0$. The range of C is contained in $\mathcal{S}^\perp$. Conversely, for any $y \in \mathcal{S}^\perp$, full row rank of A_0 gives a v with $A_0 v = y$. Then $C v = P y = y$. The range is therefore all of $\mathcal{S}^\perp$, of dimension $r - q$. The excited subspace is exactly what B_T shows us. Projecting the released A_T onto its complement gives C without the seed, and $q < r$ makes it nonzero. This is Claim 2 seen from the released factors.

5. What this gives

Theorem. Under A2–A5 and for almost every initialization in A1, $C = P_T A_T$ satisfies $C H = 0$ and $\text{rank } C = r - q$.

The test works at any finite SGD step satisfying excitation, without convergence. Each candidate gets its own test, with exactly $r - q$ independent equations on its features, and the chart turns these into a search over k image coordinates. Eight equations can isolate a candidate locally on a smooth chart with $k \leq 8$, provided their Jacobian has rank k . Larger adapter ranks and additional valid layer certificates provide more equations for higher-dimensional charts.

Mixtures. C is linear and vanishes on every private feature, so it also vanishes on their linear combinations. At the feature level, mixtures pass. The image search reaches the certificate through $h(z) = \phi(G(z))$. If $\phi \circ G$ is affine, blends in chart coordinates also pass automatically. This is why we use a nonlinear map between chart and adapter. Image blends then have to satisfy the certificate after that map, and are not accepted by linearity of C alone.

Remarks: NTK coordinates, factor fits and equation counts

The earlier NTK fit. Parameter linearization leaves a nonlinear dependence on the images. Our earlier fit instead treated B_TA_T as a raw-gradient mixture in merged-weight coordinates. With $G_0 := \nabla_W L_0 = D_0 H^\top$, Claim 1 gives the exact first-step identities

$$A_1 = A_0, \quad B_1 = -\eta_0 G_0 A_0^\top, \quad B_1 A_1 = -\eta_0 G_0 A_0^\top A_0.$$

The raw-gradient fit omits this transformation. This is a coordinate mismatch, not evidence against small-step linearization. An NTK attack can instead use gradients in the trained coordinates (A, B) ; at initialization its A -gradient is zero. This uses the initialization, whereas the following span fit uses only the released factors. The span fit permits arbitrary output-side coefficients; it does not enforce the full dependence of the loss gradients on the images.

Lemma: free-coefficient fit and its symmetry. Fix the release with rank $B_T = N$. Let $X = [h(z_1), \dots, h(z_N)]$, $V = A_T X$, and let $U \in \mathbb{R}^{m \times N}$ contain free coefficient columns. Then

$$\exists U : B_T = UV^\top \iff CX = 0 \text{ and } \text{rank } V = N.$$

Proof. An exact fit forces $\text{rank } V = N$ and $\text{row}(B_T) = \text{col}(V)$, hence $P_TV = CX = 0$. Conversely, the two right-hand conditions make $\text{col}(V)$ an N -dimensional subspace of $\text{row}(B_T)$. The spaces are equal, so each row of B_T has the required coefficients. Under A1–A5 with $q = N$, the true inputs give an exact fit provided $\text{rank}(A_TH) = N$, an additional assumption.

For every invertible $S \in \mathbb{R}^{N \times N}$,

$$U(A_TX)^\top = (US)(A_TS^{-\top})^\top.$$

Thus the *free-coefficient fit* has a change-of-basis symmetry. It is not automatically a symmetry of training replay: transformed columns must be realizable images, and their actual loss derivatives must reproduce the released factors. The columns of U need not be the accumulated backpropagation signals of the corresponding images. At a softmax head with cross-entropy, the derivative is $p_i - y_i$, where p_i is the probability vector and y_i the one-hot label. Keeping these image-dependent relations can exclude mixtures; it does not by itself prove uniqueness.

What the certificate leaves undetermined. Under A1–A5,

$$\ker C = \text{col}(H) \oplus \ker A_0, \quad \dim \ker C = q + n - r.$$

Indeed, $Cv = 0$ iff $A_0 v \in \text{col}(A_0 H)$, iff $v = Ha + w$ for some $w \in \ker A_0$. The sum is direct because A_0 is injective on $\text{col}(H)$. Therefore $\ker C = \text{col}(H)$ only when $r = n$. A nonlinear chart-to-feature map can restrict these compatible features to isolated images; nonlinearity alone does not guarantee that outcome.

Equations per image. At layer ℓ , put $q_\ell = \text{rank } H_\ell$ and $h_\ell(z) = \phi_\ell(G(z))$. A valid certificate gives $r_\ell - q_\ell$ feature equations. Write $J_\ell = \partial h_\ell / \partial z$, and let F_ℓ be the normalized residual of Section 2. At a zero with $A_{T,\ell} h_\ell(z) \neq 0$,

$$DF_\ell(z) = \frac{C_\ell J_\ell(z)}{\|A_{T,\ell} h_\ell(z)\|_2}, \quad \text{rank}(C_\ell J_\ell) \leq \min(r_\ell - q_\ell, k).$$

The denominator's derivative is multiplied by the zero numerator. Stacking valid layers gives at most $\min(k, \sum_\ell (r_\ell - q_\ell))$ first-order constraints; the actual count is the stacked Jacobian rank. Deep-layer drift and extra views can enlarge the private span. The theorem here is for fixed inputs; matrix rank alone is not a count of recoverable images.

Related work. Loo et al. (2023) study NTK reconstruction, including early stopping. Flora (Hao et al., ICML 2024) uses a frozen-initialization compressed-gradient approximation; LoRA-FA studies frozen-factor training. Our certificate follows from the exact finite-step invariant in Claim 2.